

Aircraft Fault Analysis - Minimal Report

Dataset and validation design

The workflow used 23 files and 312,501 rows after rebuilding the data from the raw Excel files. Validation is grouped by file to reduce row-level leakage. Fault onsets are provisional because they are inferred from anomaly behavior rather than externally verified labels. Direct anomaly-score columns are excluded from the supervised prediction features to reduce circularity.

Anomaly detection summary

The healthy-reference anomaly model used PCA reconstruction error and Isolation Forest scores. The selected anomaly threshold was 94.004. Healthy file false alarm rate was 0.111; defective file detection rate was 0.143.

Fault prediction summary

The fault-prediction model used causal rolling features and grouped cross-validation. Row-level ROC-AUC was 0.977, average precision was 0.603, precision was 0.435, recall was 0.500, and F1 was 0.465. File-level false alarm rate on healthy files was 0.333.

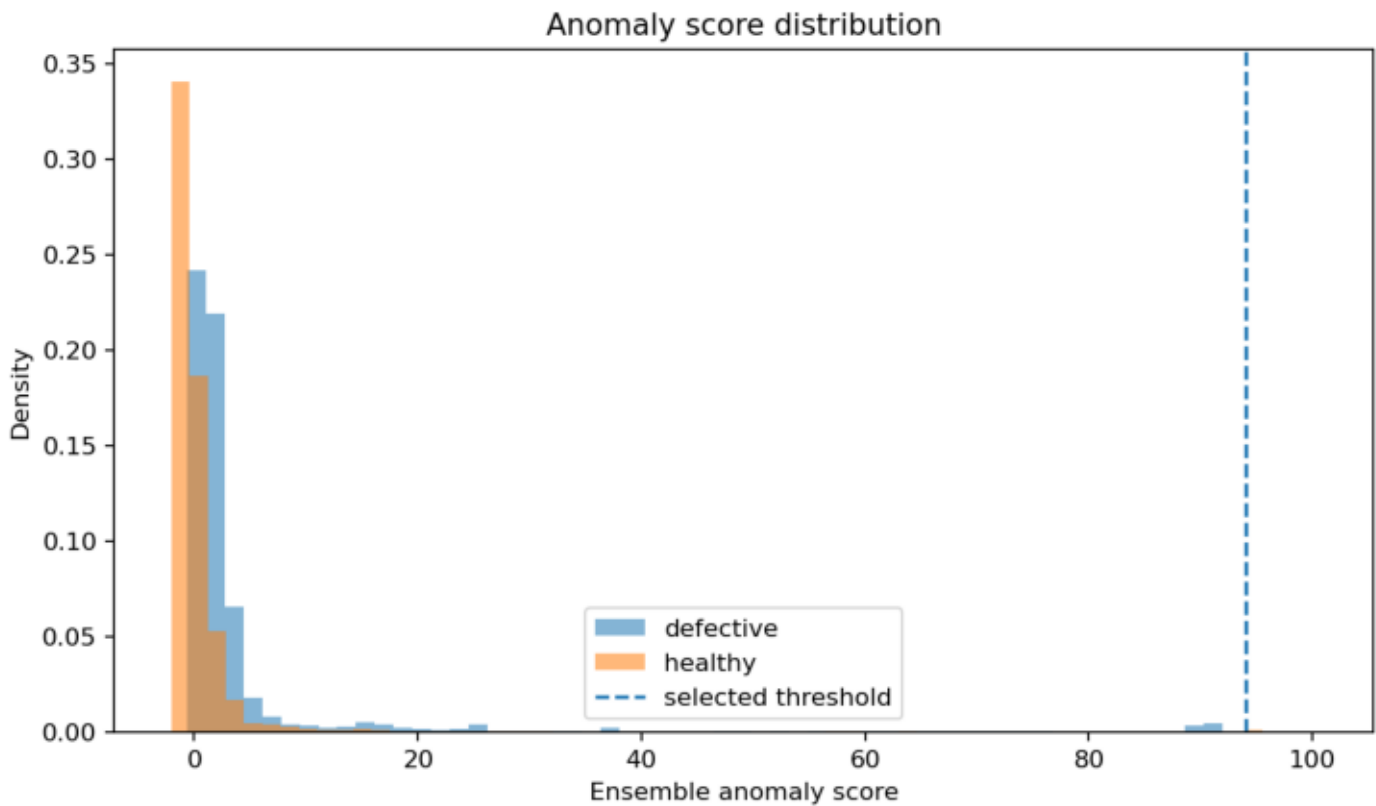
Scientific reliability note

These outputs are more realistic than treating every defective row as a clean target, but they should still be interpreted as experimental. The main limitation is that the pre-fault window is inferred from anomaly-derived onset estimates. External onset labels or manually reviewed onset points are required before making strong scientific claims.

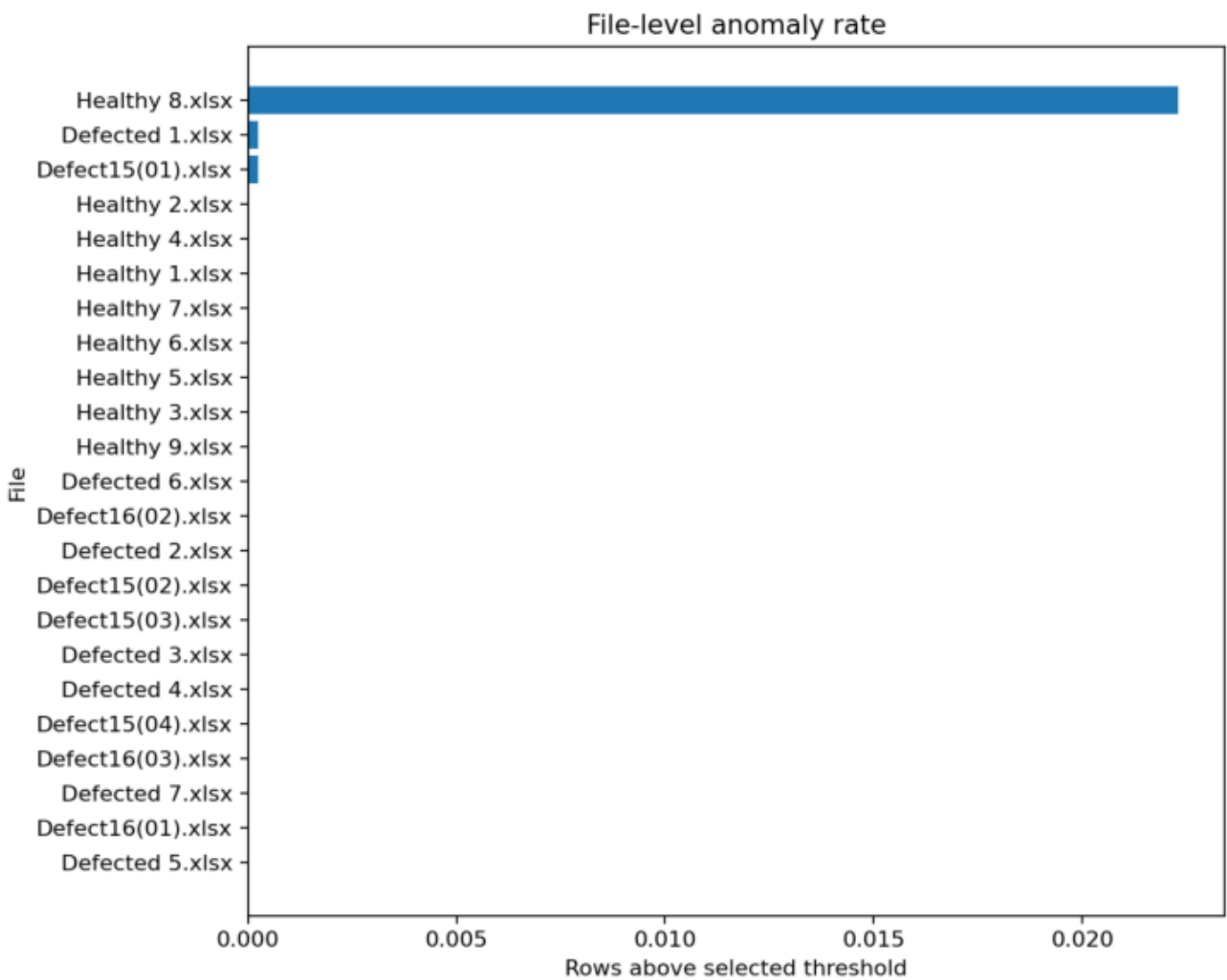
Reliability flags

high_healthy_file_false_alarm_rate: high - The selected threshold may still be too sensitive for deployment.

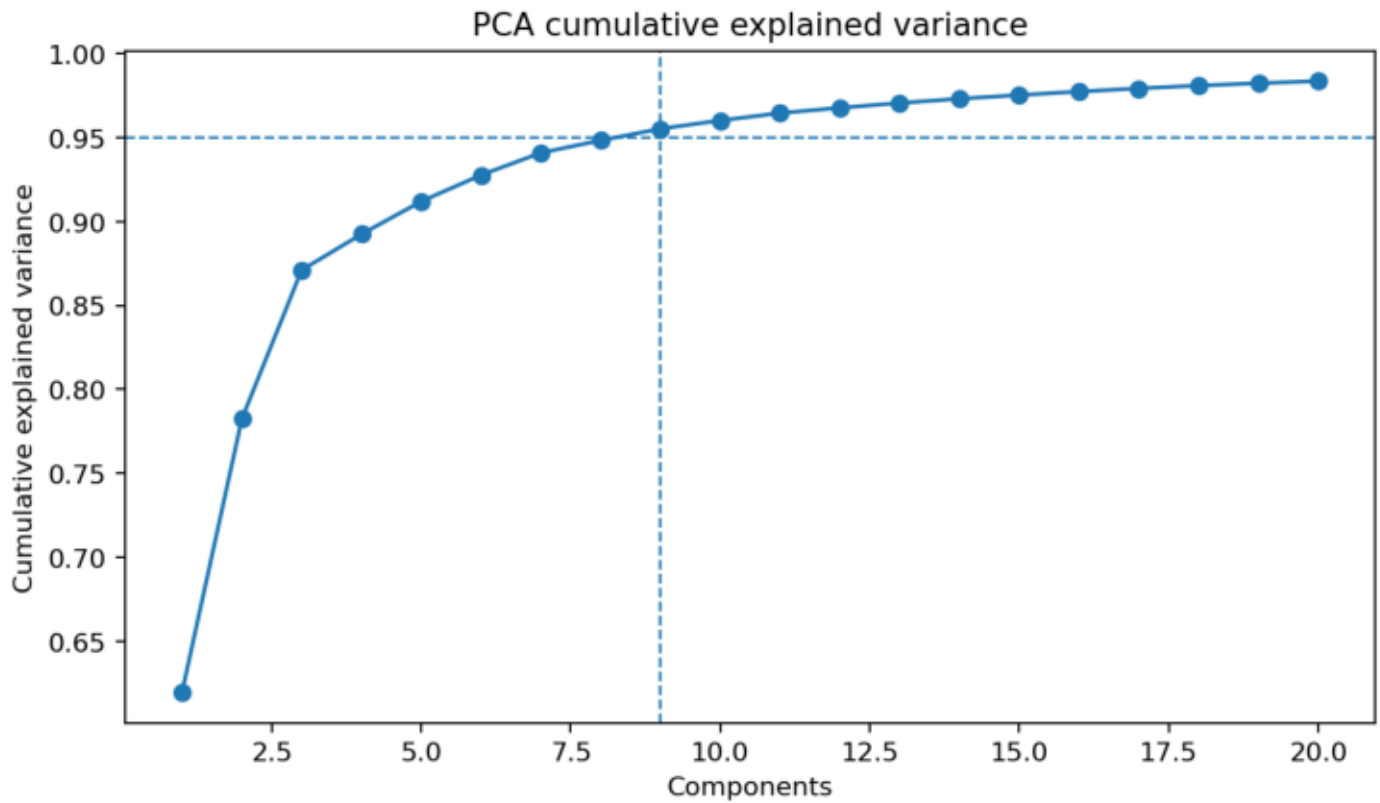
Anomaly score distribution



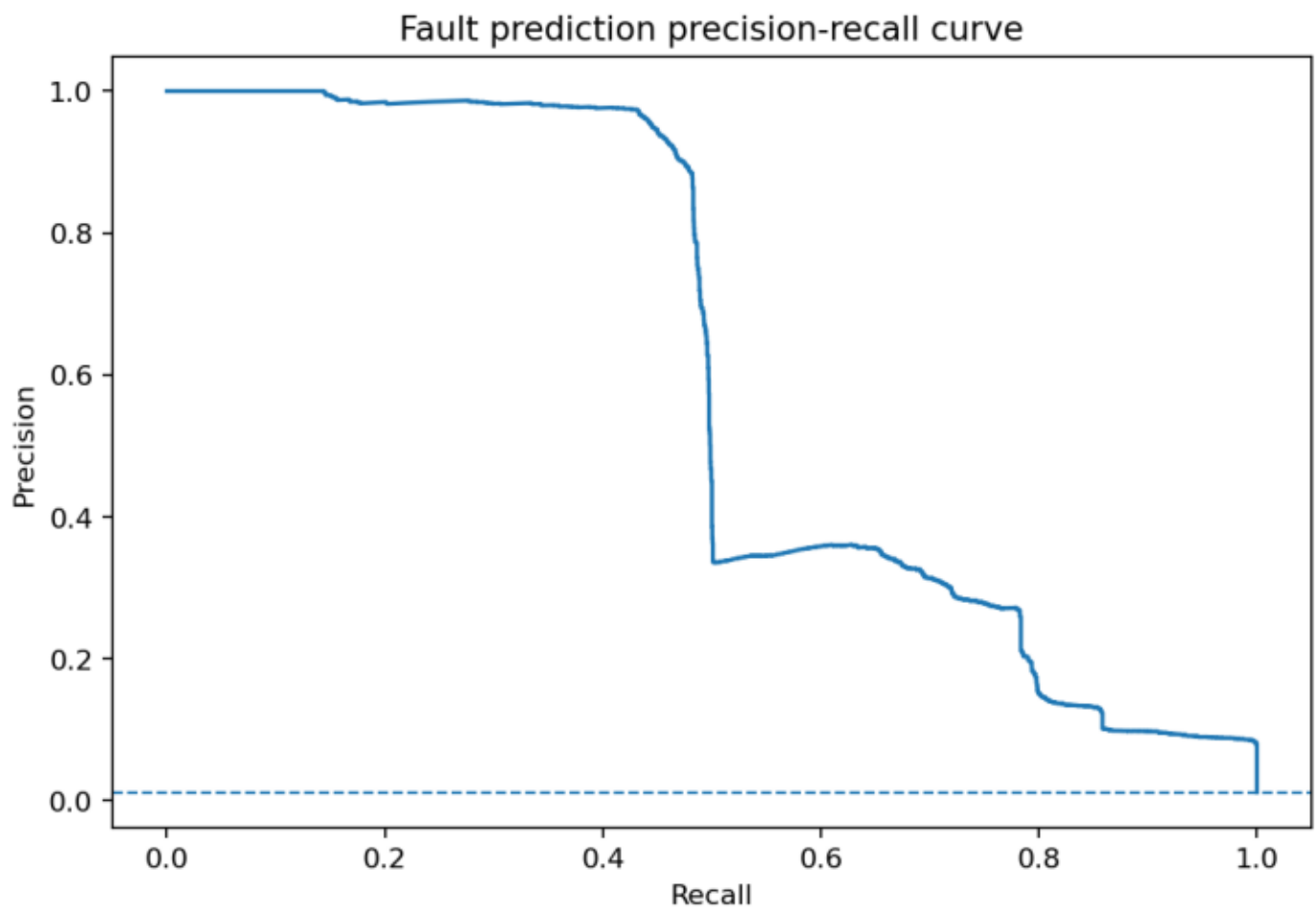
File-level anomaly rates



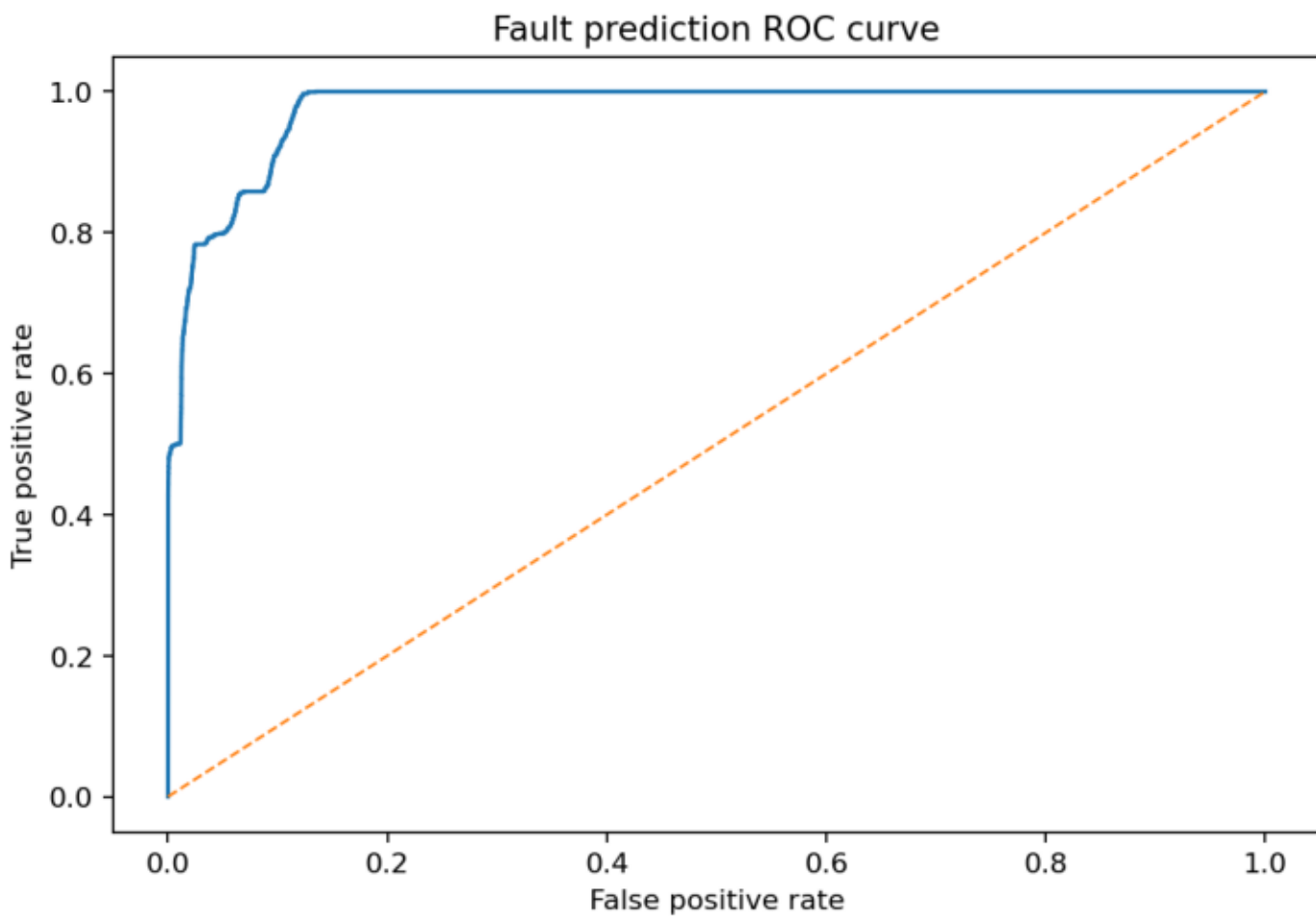
PCA explained variance



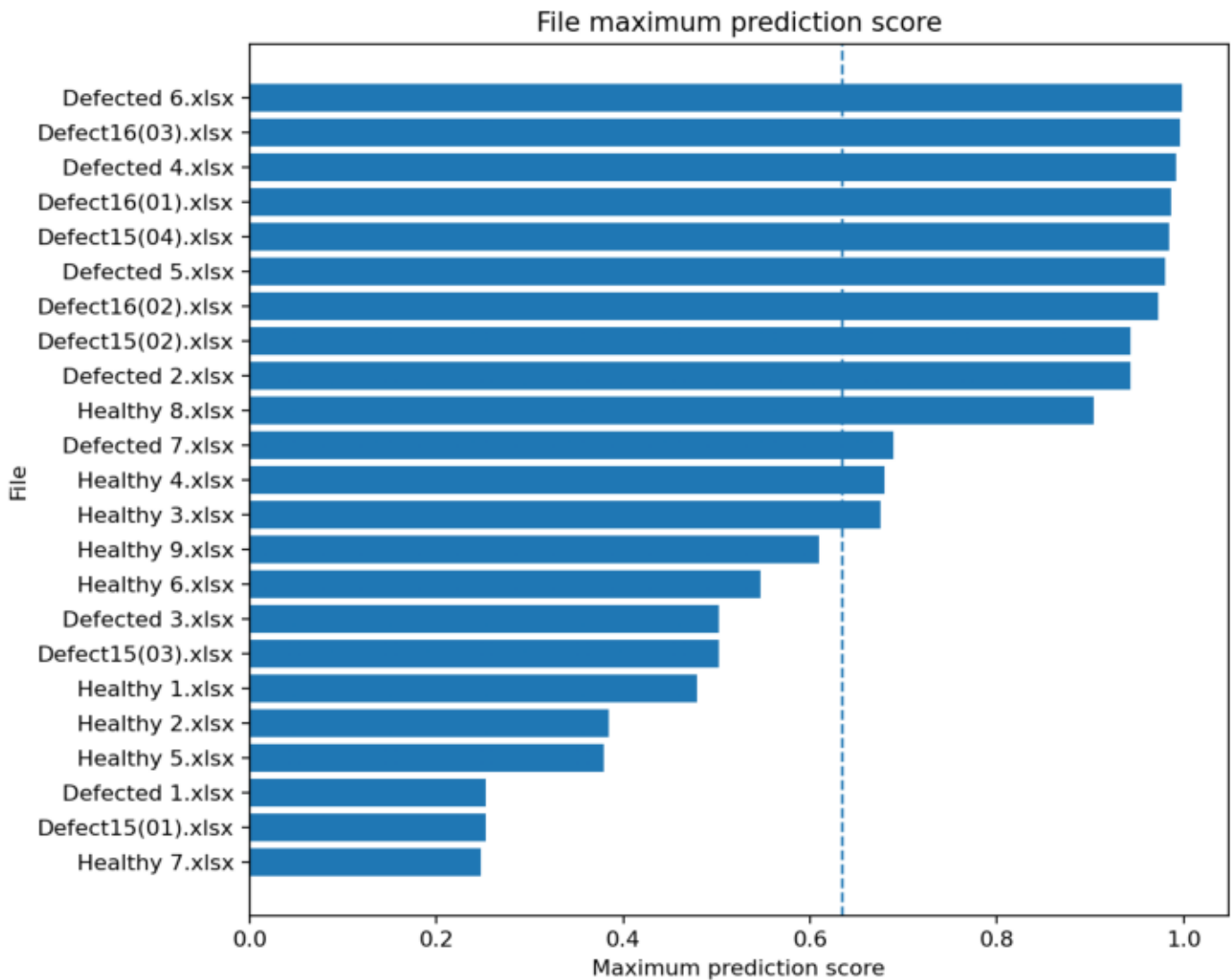
Fault prediction precision-recall curve



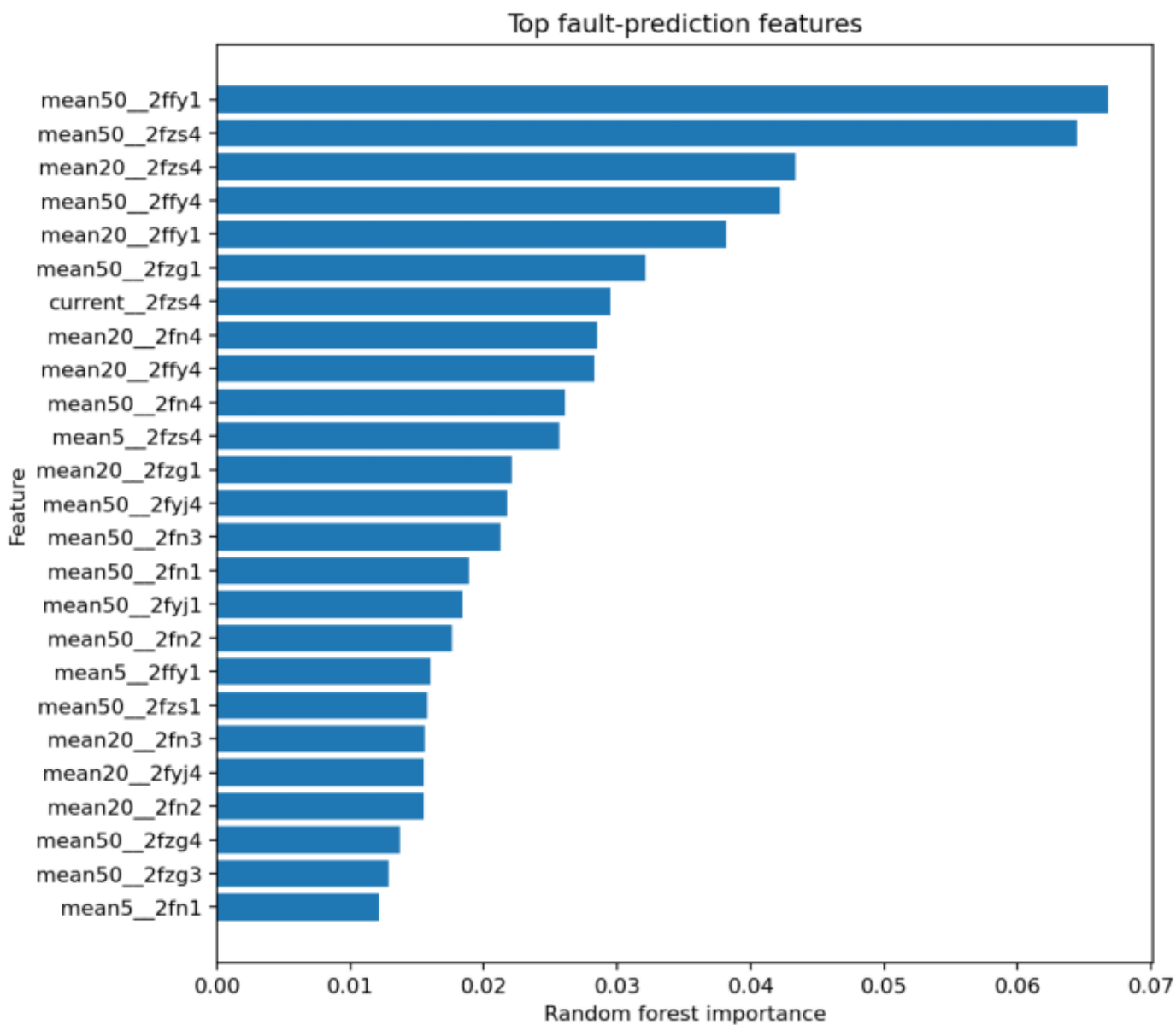
Fault prediction ROC curve



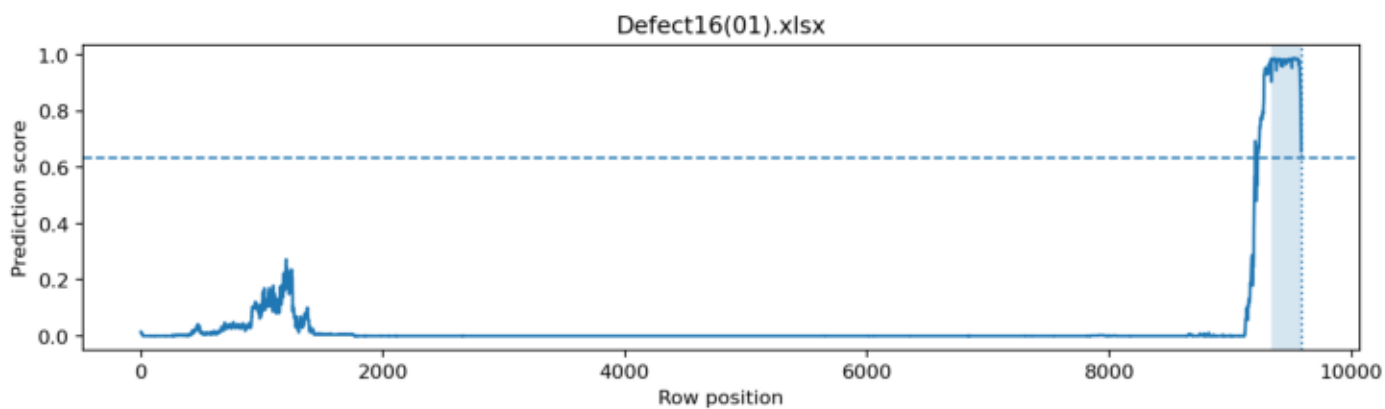
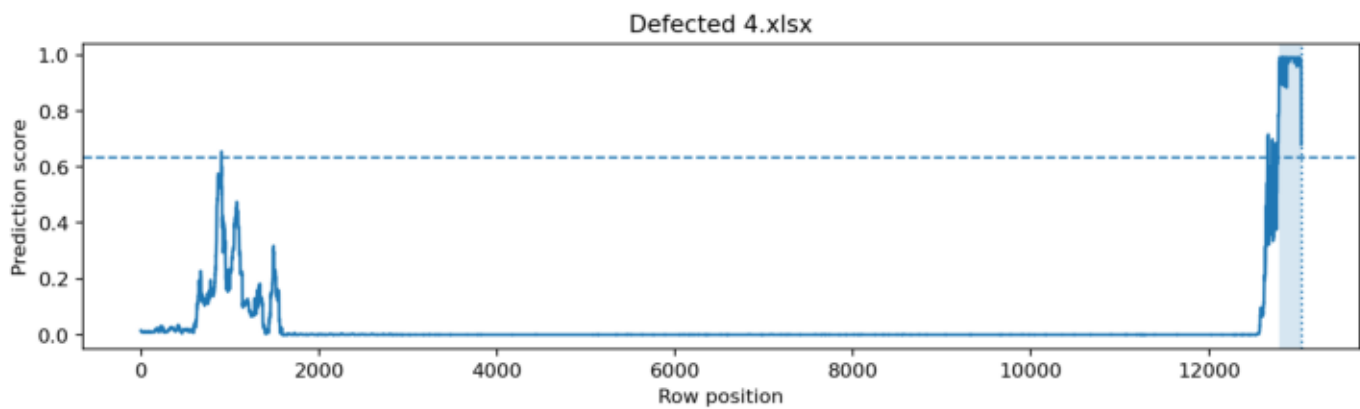
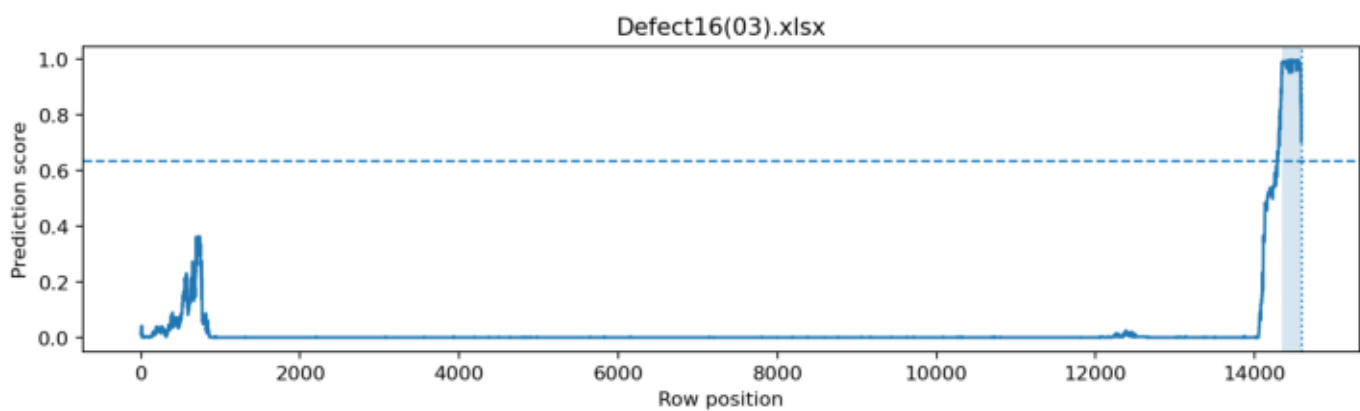
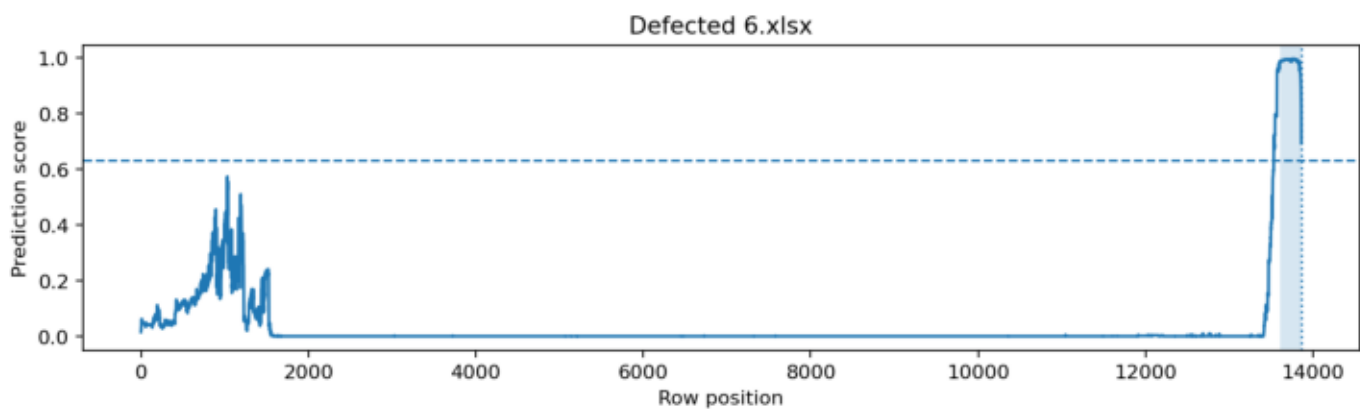
File-level prediction scores



Top predictive features



Prediction timelines



Confusion matrix

Prediction confusion matrix

